FINAL PROJECT · ADVANCED CLOUD COMPUTING

Satisfaction Meter

Emotion-Driven Personalised Marketing on AWS

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OVERVIEW

Problem & Solution

The Problem

- Generic emails ignore customer mood
- Same message for happy and frustrated customers
- Low engagement, high unsubscribe rates

Our Solution

- Customer uploads photo
- Rekognition detects dominant emotion
- Personalised SES email dispatched
- Every result stored in DynamoDB

8 CDK Stacks on AWS

Capture

S3 + presigned URL Lambda

Inference

Rekognition + EventBridge

Messaging

SES + bounce handler

Api

GET results / POST confirm

Analytics

Emotions · Campaigns · Trends

AdminAuth

Lambda Authorizer + login

Web

CloudFront + OAC + Next.js

Observability

CloudWatch dashboard

COMPUTE

15 Lambda Functions

01 Presigned URL Generator	02 Inference (Rekognition)	03 Send Email Dispatcher
04 Get Result	05 Analytics: Emotions	06 Analytics: Campaigns
07 Analytics: Trends	08 Admin Authorizer	09 Admin Login
10 Admin Submissions	11 Bounce Handler	12 Confirm Result

UPLOAD FLOW

Image Capture — Presigned URL Pattern

1 **face-api.js (browser)**

Detects face client-side. Auto-snaps after 1.5s. UX only then Rekognition classifies it

2 **POST /upload**

Lambda validates email, content-type, size. Creates DynamoDB record. Returns presigned URL.

3 **Browser → S3**

Browser PUTs image directly. Lambda never touches the bytes.

4 **EventBridge**

S3 Object Created fires to default bus. Rule routes to Inference Lambda.

Why Presigned URLs?

- Lambda never receives image bytes - no memory pressure, no timeout risk
- Browser uploads directly to S3 - lower latency, no double transfer
- CORS restricted to satisfactionmeter.live - presigned URL is the security boundary

Emotion Detection — Amazon Rekognition

1. HeadObject validates the content-type + file size (< 5 MB)
2. If invalid → delete from S3, write status: invalid_file
3. DetectFaces (Attributes: ALL) - 8 emotion scores per face
4. If no face → write status: no_face_detected
5. Sort by confidence. Tie → EMOTION_PRIORITY list
6. DynamoDB: dominantEmotion, emotionScores, status: emotion_detected

EMOTION_PRIORITY

1. happy
2. surprised
3. calm
4. neutral
5. sad
6. angry
7. fearful

Used only as a tie-breaker when two emotions share the same confidence score.

KEY INNOVATION

Two-Phase Email Pipeline

Phase 1 — Analyse

- Browser submits photo → POST /upload
- Image PUT to S3 → EventBridge fires
- Rekognition returns scores → DynamoDB updated
- UI polls GET /results/{id} until emotion appears
- Admin sees detected emotion, enters recipient email

Phase 2 — Confirm & Send

- Admin clicks Confirm → POST /confirm/{id}
- Lambda 12 sets confirmed = true in DynamoDB
- DynamoDB Stream fires → Send Email Lambda
- confirmed === true gate passes
- SES sends email, DynamoDB: status = email_sent

9 Emotion-Mapped Email Templates

HAPPY

Your smile made our day - leave us a review!

SAD

Something to brighten your day + discount code

ANGRY

We sincerely apologise - 30% off: SORRY30

SURPRISED

Flash deal just for you!

NEUTRAL

A special offer just for you

DISGUSTED

We want to hear from you

FEAR

We're here to help (key: fear, not fearful)

CALM

Exclusive member offer

CONFUSED

We noticed something unexpected - let us help

DASHBOARD

Admin Portal & Analytics

https://satisfactionmeter.live

KPI Strip

Total submissions, emails sent, top emotion - refreshes on load

Emotion Distribution

Recharts bar chart from DynamoDB scan (60s cache)

Campaigns Table

Per-template send counts with earliest / latest timestamps

Trends Chart

Daily emotion counts over 30 days (Recharts line chart)

Emotion Capture Panel

Live webcam + face-api.js overlay, Phase 1 + Phase 2 flow

Submissions Audit Log

Full table, masked emails, newest-first, status badges

Security, Infrastructure & CI/CD

Security

IAM Least-Privilege

One execution role per Lambda, scoped resource policies

SSE-S3 Encryption

AES-256 AWS-managed keys - free (KMS CMK excluded, $1/month)

SSM Parameter Store

3 SecureString params for admin creds (Secrets Manager excluded)

SQS Dead Letter Queues

3 DLQs: Inference, Messaging, BounceHandler - 14-day retention

30-Day S3 Lifecycle

Raw images deleted after 30 days. Emotion scores stored separately.

DevOps & IaC

AWS CDK (TypeScript)

8 stacks, each with its own IAM boundary

GitHub Actions + OIDC

No stored AWS credentials — short-lived JWT via federated identity

Two Workflows

deploy.yml (CDK) + frontend-deploy.yml (S3 sync + CF invalidation)

CloudFront + OAC

Private S3 bucket. CloudFront Function rewrites /login → index.html

DynamoDB — 3 Tables

Submissions (+ Stream), Campaigns, EmailSuppression

SUMMARY

Project Metrics at a Glance

15	8	9	< 5s	~$0
Lambda Functions (12 custom + 3 CDK)	CDK Stacks Deployed	Email Templates (one per emotion)	End-to-End Target Latency	AWS Spend (free tier)

Key Design Decisions

- EventBridge over direct S3-to-Lambda - avoids CDK cross-stack circular dependency
- Two-phase confirm gate - protects SES sender reputation, ensures consent
- SSM over Secrets Manager - functionally identical, saves $0.40/secret/month
- SSE-S3 over KMS CMK - AWS-managed keys are free; customer-managed keys cost $1/month

LIVE DEMO

https://satisfactionmeter.live

1. Log in to admin portal → 2. Face webcam → auto-snap → 3. View detected emotion → 4. Enter email + Confirm → 5. Check inbox

Thank You

Questions & Discussion

15 Lambda functions · 8 CDK stacks

ap-southeast-1 · Serverless · Event-driven

9 emotion templates · ~$0 AWS spend

GitHub OIDC · Two-phase email pipeline